Page a day worksheet **PAD nr: 4**

Grade 10 – Graphs and Equations – 1

Name:	Memo

QUESTION 1

Solve for x in the following equations. Show ALL calculations.

(a) $ax + b = cx + 2$ (3)

$ax - cx = 2 - b$

$x(a-c) = 2-b$

$x = \frac{2-b}{a-c}$

(b) $5^{x-1} = \left(\frac{1}{25}\right)^{x+2}$ (3)

$5^{x-1} = (5^{-2})^{x+2}$

$5^{x-1} = 5^{-2x-4}$

$x-1 = -2x-4$

$3x = -3$

$x = -1$

[6]

QUESTION 2

Given: $g(x) = \left(\frac{1}{3}\right)^x - 9$

(a) Calculate the intercepts with the axes. (4)

y-int (put $x=0$)

$y = (\frac{1}{3})^0 - 9$

$y = 1 - 9$

$y = -8$

$(0;-8)$

x-int (Put $y=0$)

$0 = (\frac{1}{3})^x - 9$

$9 = (\frac{1}{3})^x$

$9 = (3^{-1})^x$

$9 = 3^{-x}$

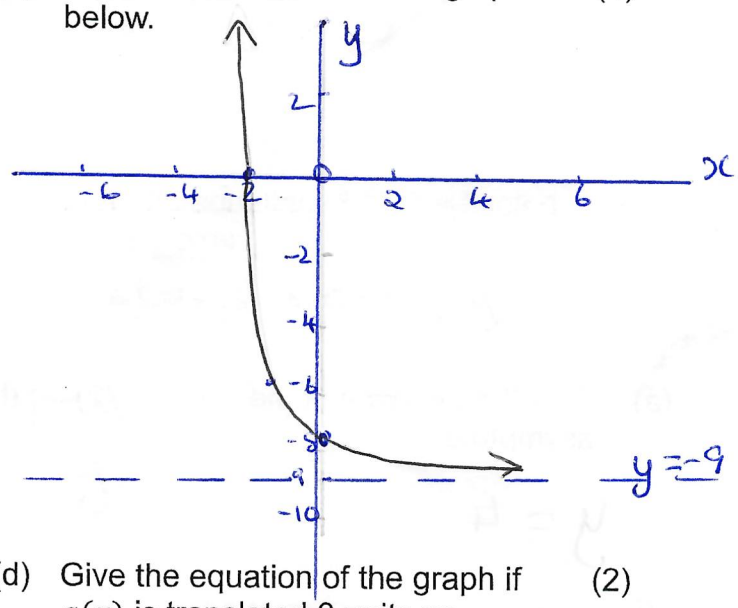
$3^2 = 3^{-x}$

$2 = -x$

$-2 = x$

$(-2;0)$

(b) Make a neat sketch of the graph below. (4)



(c) Give the range of the graph. (2)

$y > -9$

OR

$y \in (-9; \infty)$

(d) Give the equation of the graph if $g(x)$ is translated 3 units up. (2)

$y = (\frac{1}{3})^x - 9 + 3$

$y = (\frac{1}{3})^x - 6$

[12]

Page a day worksheet **PAD nr: 5**

Grade 10 – Graphs and Equations – 2

Name:	

QUESTION 1

Solve for x in the following equations. Show ALL calculations.

(a) $(x-2)(2x+5)=0$ (2)

$x-2=0$ or $2x+5=0$

$x=2$ or $2x=-5$

$x=-5/2$

(b) $0{,}125-\left(\frac{1}{2}\right)^{x+2}=0$ (4)

$\frac{1}{8}-\left(\frac{1}{2}\right)^{x+2}=0$

$\frac{1}{2^3}=\left(\frac{1}{2}\right)^{x+2}$

$\left(\frac{1}{2}\right)^3=\left(\frac{1}{2}\right)^{x+2}$

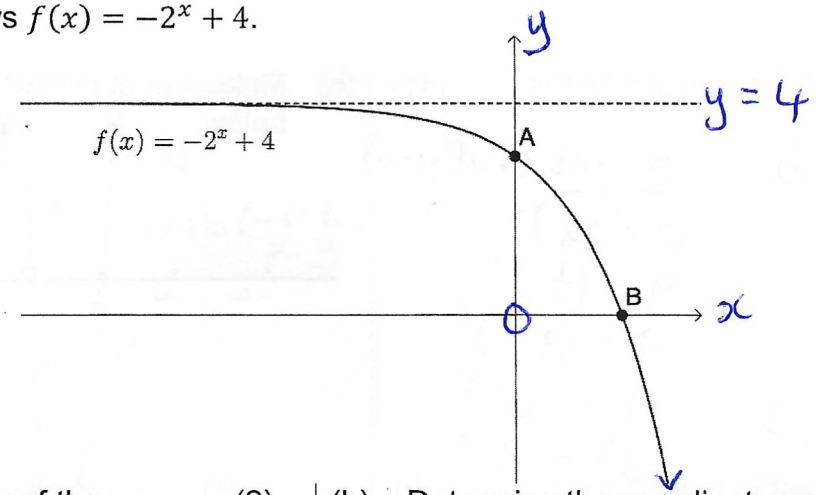
$3=x+2$

$1=x$

[5]

QUESTION 2

The graph below shows $f(x)=-2^x+4$.



(a) Give the equation of the asymptote. (2)

$y=4$

(b) Determine the coordinates of A and B. (3)

y-int (Put $x=0$)

$y=-2^0+4$

$y=-1+4$

$y=3$

$(0;3)$

x-int (Put $y=0$)

$0=-2^x+4$

$2^x=4$

$2^x=2^2$

$x=2$

$(2;0)$

(c) Give the equation reflection of the graph in the y-axis. (2)

"y stays same, x-changes sign"

$y=-2^{-x}+4$

$y=-(2^{-1})^x+4$

$y=-\left(\frac{1}{2}\right)^x+4$

(d) Give the range of the graph. (2)

$y<4$

OR

$y\in(-\infty;4)$

[9]

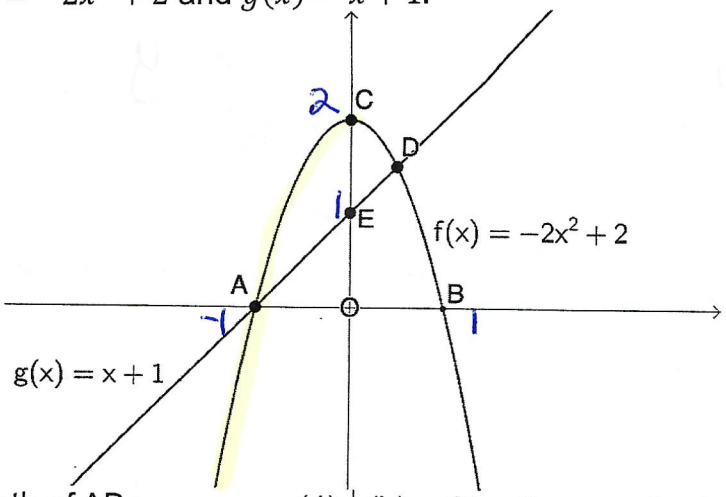
ge a day worksheet **PAD nr: 6**

rade 10 – Graphs and Equations – 3

Name:	

QUESTION 1

The graph shows $f(x) = -2x^2 + 2$ and $g(x) = x + 1$.



(a) Determine the length of AB. (4)

$-2x^2 + 2 = 0$ (Find x intercepts)

$-2(x^2 - 1) = 0$

$-2(x+1)(x-1) = 0$

$\therefore x = -1$ or $x = 1$

at A at B

$\therefore$ AB = 2 units

(b) Give the coordinates of E. (2)

E is y-intercept of the straight line g(x)

$\therefore$ E (0 ; 1)

(c) Give the range of $f(x)$ (2)

$y \leq 2$

OR

$y \in (-\infty ; 2]$

(d) Give the equation of $f(x)$ reflected in the x-axis. (2)

"x stays same, y changes sign"

$-y = -2x^2 + 2$

$\therefore y = 2x^2 - 2$

(e) For which values of x will $f(x)$ be increasing? (2)

See highlighted section of graph

$\therefore x < 0$

(f) Describe the transformation of $g(x)$ to $y = -x + 1$ (2)

"y stayed same, x changed sign"

$\therefore$ reflected in y-axis

[14]

Page a day worksheet **PAD nr: 7**

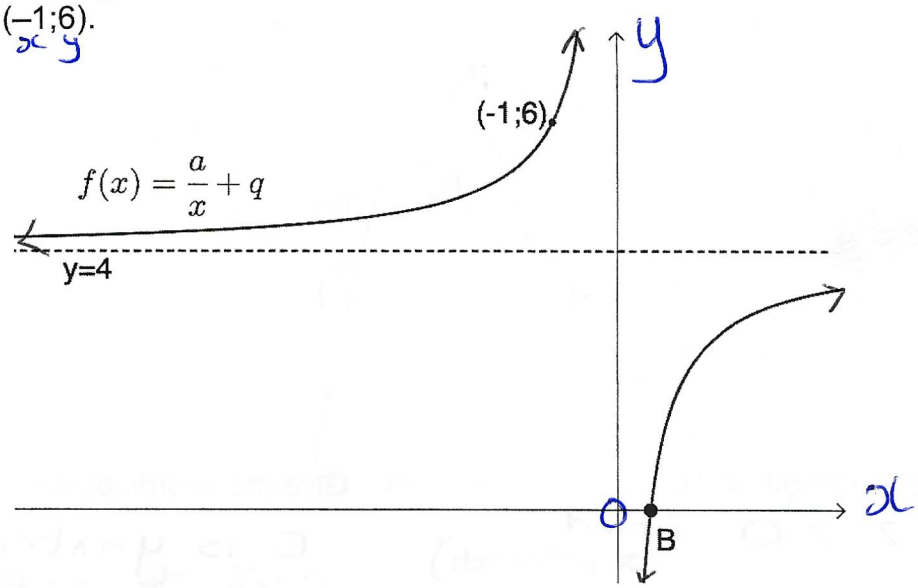
Grade 10 – Graphs and Equations – 4

Name:	

QUESTION 1

The graph shows $f(x) = \frac{a}{x} + q$ which has a horizontal asymptote at $y = 4$ and passes through the coordinates of (–1;6).

x y



(a) Prove that the values of $a = -2$ and $q = 4$. (4)

Sub asymptote $y = 4$

$\therefore f(x) = \frac{a}{x} + 4$ sub (-1; 6) (x y)

$6 = \frac{a}{-1} + 4 \rightarrow q$

$2 = -a$

$-2 = a$

(b) Determine the coordinates of B. (3)

B is x-intercept (put $y = 0$)

$y = \frac{-2}{x} + 4$

$0 = -\frac{2}{x} + 4$

$\frac{2}{x} = 4$

$2 = 4x \rightarrow \frac{1}{2} = x$

$B(\frac{1}{2} : 0)$

(c) Give the domain and range of $f(x)$. (2)

Domain: $x \in R, x \neq 0$

Range: $y \in R : y \neq 4$

(d) Give the equation of the axis of symmetry with a positive gradient. (2)

translations must apply

$\therefore y = x + q$

$y = x + 4$

(e) Determine the equation of the reflection of $f(x)$ in the y-axis. (2)

y-stays same, x changes sign

$y = \frac{-2}{-x} + 4$

$y = \frac{2}{x} + 4$

[14]